

Evidence for decreasing quality of semen during past 50 years.

OBJECTIVE: To investigate whether semen quality has changed during the past 50 years. DESIGN: Review of publications on semen quality in men without a history of infertility selected by means of Cumulated Index Medicus and Current List (1930-1965) and MEDLINE Silver Platter database (1966-August 1991). SUBJECTS: 14,947 men included in a total of 61 papers published between 1938 and 1991. MAIN OUTCOME MEASURES: Mean sperm density and mean seminal volume. RESULTS: Linear regression of data weighted by number of men in each study showed a significant decrease in mean sperm count from $113 \times 10^6/\text{ml}$ in 1940 to $66 \times 10^6/\text{ml}$ in 1990 ($p < 0.0001$) and in seminal volume from 3.40 ml to 2.75 ml ($p = 0.027$), indicating an even more pronounced decrease in sperm production than expressed by the decline in sperm density. CONCLUSIONS: There has been a genuine decline in semen quality over the past 50 years. As male fertility is to some extent correlated with sperm count the results may reflect an overall reduction in male fertility. The biological significance of these changes is emphasised by a concomitant increase in the incidence of genitourinary abnormalities such as testicular cancer and possibly also cryptorchidism and hypospadias, suggesting a growing impact of factors with serious effects on male gonadal function.